IMAAN GILL

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EDUCATION

Toronto Metropolitan University Toronto, ON

Bachelor of Engineering, Computer Engineering Expected Apr 2029

Relevant Coursework: Digital Systems, Software Systems, Engineering Algorithms & Data Structures, Object Oriented Engineering Analysis & Design, Electric Networks, Electronic Circuits I

TECHNICAL SKILLS

Programming: C, C++, Java, Python, HTML/CSS, VHDL, Verilog
Embedded & Hardware: Arduino, ESP32, FPGA/CPLD, digital logic, microcontrollers, breadboarding, soldering, 3D printing, HCS12
Electronics & Lab: KiCad, NI Multisim, Oscilloscopes, Function generators, DMMs, MOSFET/BJT circuits, Circuit Prototyping
Software & Tools: Quartus Prime, ModelSim, MATLAB, NetBeans, JavaFX, VS Code, JUnit, UML, Ant, Maven, Gradle, Assembly, Microsoft Office Suite

ENGINEERING PROJECTS

8 Bit FPGA General-Purpose Processor Dec 2025

VHDL, Quartus Prime, FPGA, FSMs, Digital Logic

- Designed and verified an 8 bit FPGA testing board with registers, FSM control, decoders, ALU's, and a segment display. implemented VHDL arithmetic, logic, shift and rotate, comparison, and equivalent operations, verified using waveform simulation and testing in Quartus software.

EEbot Robot Roaming Controller Dec 2026

HCS12 Assembly, CodeWarrior, FSMs, Timer Interrupts, ADC, LCD

- Programmed a six state HCS12 controller for the free roaming bot, using bumper/stern inputs and timer interrupts to control forward motion, reversing, turns, and stopping.
- Integrated the DC motor control, ADC battery voltage monitoring, and an LCD display of current battery voltage and state.

Book Store Management Application Aug 2026

Java, JavaFX, NetBeans, Ant, OOP, State Pattern, File IO

- Created a JavaFX bookstore application with owner and customer authentication, inventory management, purchasing, loyalty point redemption, and continuous file storage. Applied a User hierarchy and State pattern for customer status along with various OOP concepts.

Transistor-Level Circuit Design & Simulation 2026

KiCad, SPICE, NMOS/PMOS, Analog Electronics

- Designed and simulated a three stage 3.3 V low voltage amplifier in KiCad using NMOS, PMOS and CMOS models, resistor bias networks, and capacitor coupling. Transistor sizing and device biasing using SPICE point analysis and created the physical circuit on a breadboard.

EXPERIENCE

City of Brampton: STEAM Instructor/Facility Staff Jun 2024 - Present

Collaborative Learning & Technology Centre (CLTC), Brampton, ON

- Teach groups of 10 to 20 students ages 8 to 14 across 15+ STEM programs, covering topics such as Arduino and VEX robotics, Python, C++, OOP Python, HTML, CSS, Blender, and related engineering concepts.
- Create lesson plans and activities then build, test, and debug hardware and software with students, including VEX vehicles and programming projects.
- Create and present clear instruction plans while leading classroom problem solving, collaboration in a team environment, and safe use of electronics and robotics equipment.

LEADERSHIP, AWARDS & CERTIFICATIONS

DECA Ontario - Marketing Competitor: Advanced from regional competition to provincials while achieving Top 10 and Top 5 placements in marketing case study categories.
Certifications: WHMIS Certificate, Emergency First Aid with CPRB
Languages: English (fluent), Punjabi (conversational)